6 REGISTERS - LISTS

6.1 Theoretical part

6.1.1 Registrars

Registers are digital circuits in which information is stored. Usually their length is equal to that of a digital word. They consist of flip-flops whose number is equal to the number of binary digits that make up the word. The information recorded in the flip-flops remains in them and can be used later at any time, as long as no new change is caused in the state of the flip-flop outputs. The flip-flops used are R–S, D and J–K depending on the purpose of each application. The registers are

manufactured in integrated circuits with a length of 8, 16, 32, 64 up to 128 bits depending on the computer CPU they will support. The writing of a word is done simultaneously in the flip-flops by applying the same clock pulse to all, after applying the appropriate logic levels to their inputs.

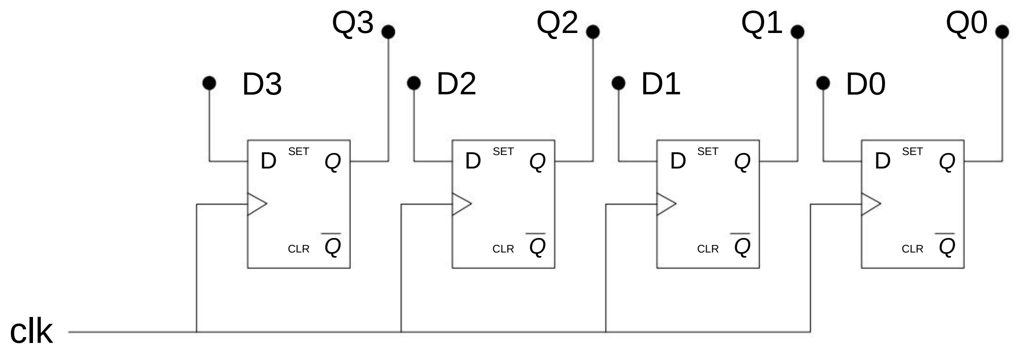


Fig. 1 Four-bit register with D f/f

Figure 1 shows a four-bit register with a D flip–flop. If we have applied logic '1' to inputs D3 and D1 and logic '0' to D2 and D0, what four-bit word will be registered after one clock pulse?

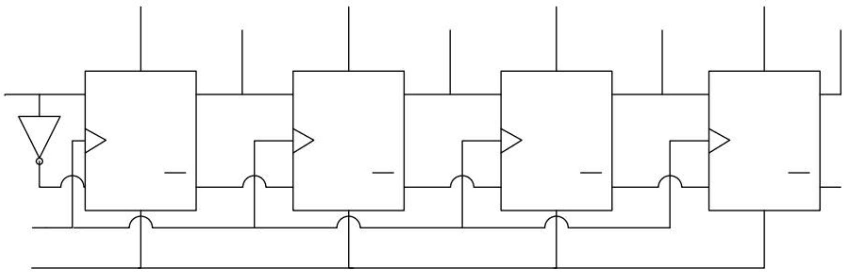
6.1.2 Sliders

Shift registers are registers whose flip-flops are connected together (outputs with inputs) in such a way that the contents of the register can be shifted right or left (depending on the connection) by applying clock pulses. Depending on the direction of movement, we have right shift registers, left

shift registers and bidirectional registers where their contents can be shifted either right or left.

In a right shift register with each shift in most significant bit can be entered as a logical '1' or a logical '0'. In case where a logical '0' is entered, the shift to the right is equivalent to division by two. E.g. if we have the number 1100 and we have a right shift by inserting '0' in the most significant digit after a shift we will get the number 0110 which is the quotient that results if the previous one divided by two.

In the case of the left shift register with each In shift we have a logical '1' or '0' inserted into the least significant digit. In case we have for each slip input a logical '0' the number is the previous one times two. For example, if we have the number 0111 with a left shift and insertion of a logical '0' in the least significant bit digit we will get the number 1110 which is actually the product of previous times two.



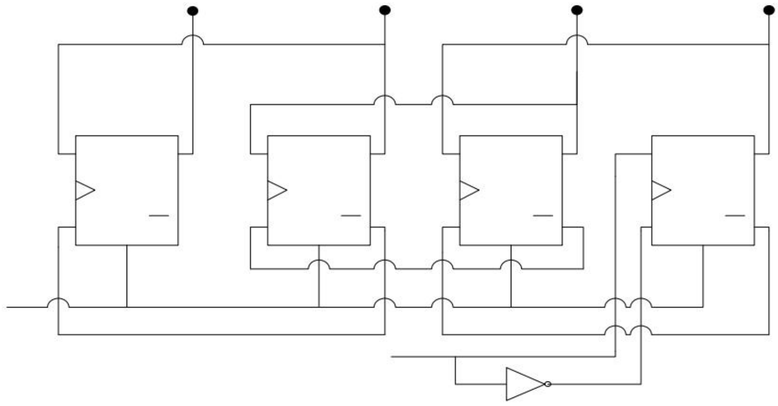
Clear	**Pulse clock**	**i/p**	**Q3**	**Q2**	**Q1**	**Q0**
1		X	0	0	0	0
0	1st	1	1	0	0	0
0	2nd	0	0	1	0	0
0	3rd	1	1	0	1	0
0	4th	1	1	1	0	1

Fig. 2 Right shift register

In figure 2 we have a four-bit right shift register implemented with 4 JK flip-flops where we see how loading can be done of the number 1101 after applying four clock pulses.

Correspondingly in figure 3 with a different connection we have a left shift register. Here it seems that the loading the same number 1101 with 4 clock pulses into a register left slip.

We observe that the right-sliding slider has one input via whose word is loaded serially starting from the least important bit of while correspondingly the left shift slider has one input through whose word is loaded serially starting from the most important bit of it.



Clear	Pulse clock	i/p	Q3	Q2	Q1	Q0
1		X	0	0	0	0
0	1st	1	0	0	0	1
0	2nd	1	0	0	1	1
0	3rd	0	0	1	1	0
0	4th	1	1	1	0	1

Fig. 3 Left shift register

In general, information in a shift register can be entered serial or parallel. Thus we have serial input - serial output (SISO - serial input serial output), serial input - parallel output (SIPO - serial input parallel output), parallel input - serial output (PISO - parallel input serial output) and parallel input registers.
- parallel output (PIPO - parallel input parallel output).

6.1.3 The 74194 integrated circuit - general purpose shift register use

The 74194 is a four-bit general purpose parallel and serial left or right load shift register.
(figure 4). Inputs S0 and S1 determine its operating mode based on table 1 and essentially the next state of the outputs QA , QB , QC , QD after a clock pulse.

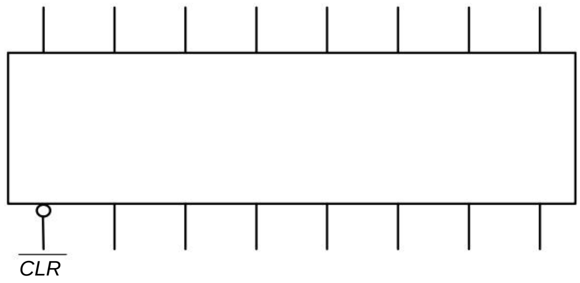


Fig. 4 Integrated 74LS194 - general purpose shift register

S1	S0	Operation
0	0	Unchanged output
0	1	Right Slip
1	0	Left Slip
1	1	Parallel input 4 Bit

Table 1 Operating modes of the 74LS194

The inputs IR, IL are the inputs for the right and left slide respectively, while inputs A,B,C and D are the data inputs for the parallel loading mode. Applying low to the *CLR* input makes reset on outputs QA,QB,QC and QD.

Using 2, 4, 8 etc. 74194 we can compose registers - shifters with length 8, 16, 32 etc. bits.

6.2 Work Execution

6.2.1 Right shift register

Connect the right register circuit below. with 4 Flip-Flops 7476 (2 integrated) and a 7404 inverter (figure 5). Connect the parallel data i/p inputs to switches (+5V, 0V), the clear, clock to button active low and the right IR slide input to switch (+5V, 0V).

a) SIPO operation.

With the parallel inputs at logic '1' (why?) and with 4 pulses clock by giving the appropriate values to the IR input before each pulse serially load the number 1001. Notice how it appears finally to the LEDs connected to the outputs of our register.

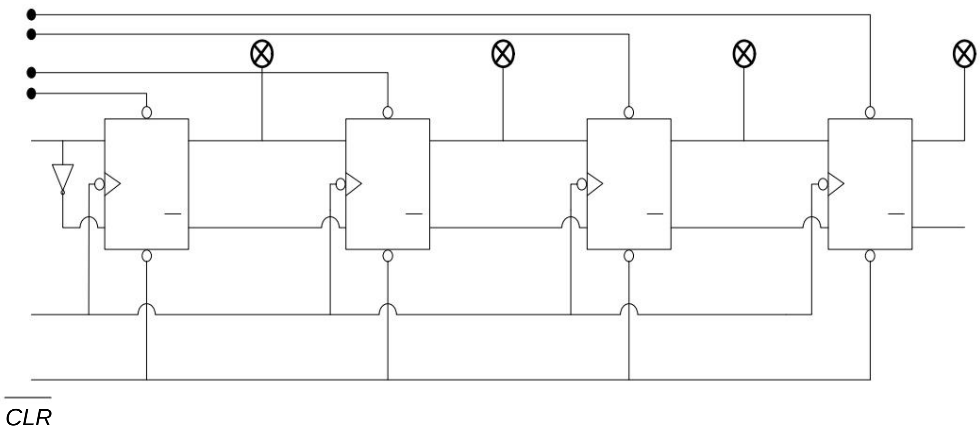


Fig. 5 Right shift register

b) PISO operation. Give

a low pulse to the *C LR* input of the register, making all outputs 0. Then give the appropriate voltages (0 or 5V) to the asynchronous parallel data inputs so that the number 0110 is loaded in parallel. Does the digital word 0110 appear on the LEDs? After resetting all parallel inputs to logical '1' (why?) show how and with how many clock pulses, we will get the 4 bits of the four-bit word 0110 in serial output from Q0 , starting from its least significant digit.

6.2.2 General purpose shift register 74194

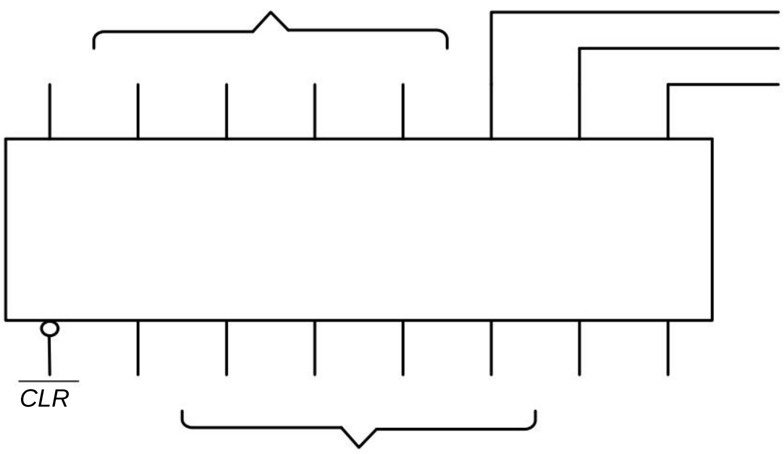


Fig. 6 General purpose slider 74LS194

Connect the outputs of the 74LS194 general purpose slider (figure 6) to LEDs. The *CLR* input to button active low and the CLK to button active high. The S1, S0 (mode) inputs, the IR, IL (right, left shift) inputs and the parallel data inputs A, B, C, D to switches (0V, +5V).

1. Set S1, S0 to logic '1' where we have parallel register loading mode (table 1) and load the binary number 1011 into the register through the
A, B, C and D parallel inputs by applying a clock pulse. Then give a low pulse to clear making all outputs zero.

2. Set S1 to logic '0' and S0 to logic '1' where we have right shift operation (table 1) and serially load the binary number 1101 through the IR input (right shift) with four clock pulses. Observe the outputs of the QA, QB, QC and QD registers on the LEDs after each clock pulse. 3. Repeat the previous paragraph for left shift with S1
to logic '1' and S0 to logic '0' (table 1) through the IL input (left shift).

4. What happens when S1 and S0 are at logic '1' and we give a clock pulse?

6.2.3 Eight-bit register and information transfer from four-bit to four-bit register

Connect the circuit in Figure 7 with two 74LS194, two AND gates, one OR gate and one inverter.

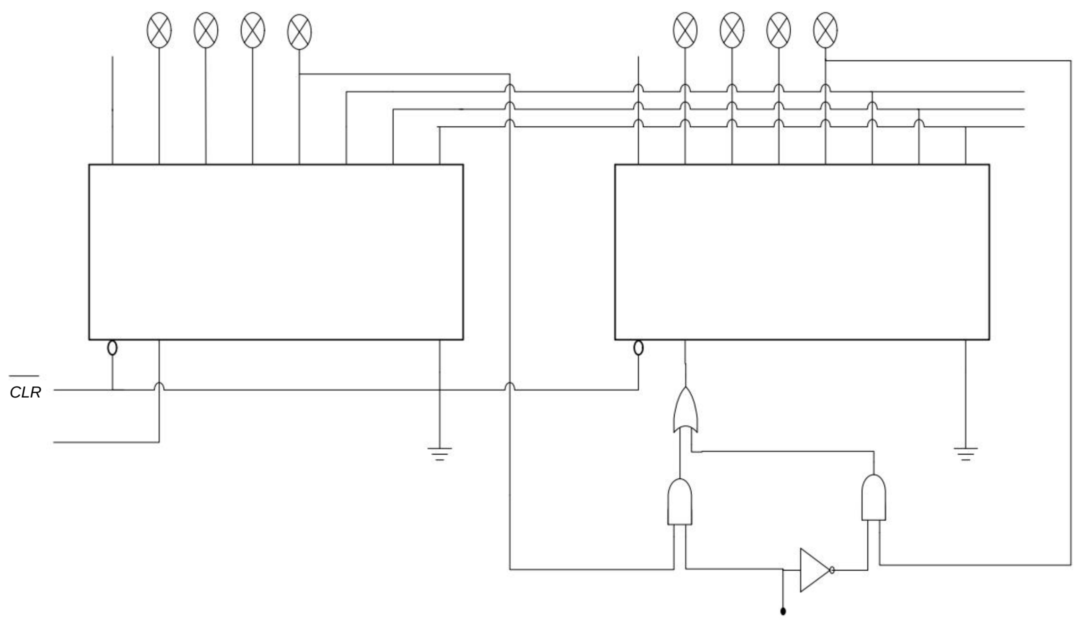


Fig. 7 Eight-bit register with control circuit

First, give a low pulse to the $\overline{CLR}$ inputs , resetting the register outputs.

a) Set the appropriate voltages at the inputs S1, S0 to obtain a right shift (table 1). With the input M at +5V, first enter the digital word 1001 into the first integrated circuit with 4 clock pulses via the IR input. Then enter the word 1101 into the first register, observing the outputs of the second. After 4 clock pulses, the word 11011001 should be entered into the two registers, which thus function as one eight-bit register. b) Then connect the input M to 0V and enter the word 1011 into the first register again with

a right shift with 4 clock pulses. What do you now observe at the outputs of the two registers? What happened to the second register? Explain.